DEN: A New Method for SAR and Optical Image Fusion and Intelligent Classification

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***Abstract*— Synthetic aperture radar (SAR) and optical images possess complementary strengths, offering rich spatial and spectral information. The intelligent classification of features through image fusion of SAR and optical presents both opportunities and challenges. However, fusion and intelligent classification encounter hurdles. Different physical properties and imaging principles between SAR and optical images often lead to sensor property mismatches, causing information loss. Moreover, optical images are susceptible to weather conditions, while SAR images suffer from scattering noise interference. In addition, the nonuniform distribution of feature categories results in sample imbalance. To address these problems, this article proposed a new fusion network structure dual-encoder net (DEN). First, without increasing the model complexity, considering the differences in the performance of features under different sensors, this network was keyed to a composition of two encoders that were able to utilize their respective features to encode and reduce the impact of modal differences. Second, a detail attention module (DAM) was constructed to capture the detailed information that was obscured by the optical image and acquired by the SAR image. Finally, a new loss function, comprising weighted information loss, pixel loss, and noise loss, was introduced to mitigate sample imbalance, retain key information, and reduce noise effects. The experimental results showed that the proposed method outperforms the current popular image fusion methods, and the model complexity was improved by 15.3% while the overall accuracy (OA) was improved by 2.6%, and the entropy, peak signal-to-noise ratio (PSNR), and mean square error (MSE) were improved by 29%, 27%, and 7%, respectively.**

***Index Terms*— Attention module, deep learning, image fusion, loss function, optical images, synthetic aperture radar (SAR) images.**

I. Introduction

MULTISOURCE image fusion involves integrating images taken at the same time, location, and angle into a single image using a specific fusion method [1]. This process minimizes redundant information and results in a fused image that is more complete, reliable, and information-rich than the individual source images [2]. Image fusion is crucial for image enhancement, sharpening, information acquisition, target detection, and recognition. Multisource image fusion techniques have been widely applied in various fields [3], including medical imaging analysis [4], [5], [6], [7], photographic and remote sensing mapping [8], [9], [10], [11], robotic vision [12], [13], surveillance and tracking [14], [15], intelligent driving [16], environmental monitoring [17], [18], [19], and land-use classification [20], [21].

Synthetic aperture radar (SAR) images essentially reflect electromagnetic scattering from features, but are heavily influenced by speckle noise and lack color and texture, insufficient for discriminating feature categories [22]. In contrast, optical images offer excellent color and texture detail information. By fusing SAR images with color and texture detail information from optical images, we can create high-quality fusion images that capitalize on the complementary information from both modalities. Moreover, optical remote sensing images often suffer from loss due to cloud cover or cloud shadows, severely limiting their utility in military and scientific research [23]. SAR images operate around the clock and possess strong penetrating capabilities in cloud-covered environments. Therefore, fusing SAR and optical images can effectively compensate for information deficiencies under cloud cover [24]. In conclusion, the fusion of optical and SAR images holds significant value as it leverages their respective strengths, extracts unique features, and produces high-quality images better suited to the needs of practical applications and scientific research.

With the development of deep learning, a lot of research on image fusion has appeared [25], [26], [27], [28], [29], [30], [31], despite existing some research in this field, the fusion of SAR and optical images remains essential due to their distinctive characteristics. Deep learning-based image fusion and intelligent classification rely on establishing connections between source and fused images through specific rules. By leveraging extensive training data and loss functions, machines learn these rules, enabling the fusion classification of new test data to achieve desired outcomes. To enhance the effectiveness of intelligent classification, research focuses on three main areas: various network structures [32], [33], [34], [35], [36] are often devised, incorporating specific attention modules [37] and constructing suitable loss functions [38].

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1) *Network Structures:* Network structures in deep learning are typically classified into two categories: simple and complex. The former boasts fewer parameters, a straightforward model structure, and faster operation. For example, Zhang et al. [39] devised IFCNN, which uses two convolutional layers to extract image features from multiple input images. Li and Wu [40] designed a simple network structure that utilizes two fusion layers in order to obtain more useful features. Although the simple network is fast, the fusion effect is often unsatisfactory. Consequently, more complex network structures have emerged, offering superior operation and feature extraction capabilities. Several representative works are as follows. Ma et al. [41] use a game of generators and discriminators for better information retention. Deng and Lv [42] proposed the SFGAN, learning the mapping relationship from small-scale SAR to optical image pairs. Yet, complex networks may lead to overfitting as depth increases. Therefore, we analyze the structure of complex networks for SAR and optical image fusion, focusing on extracting high- and low-order features to retain as much information as possible. Lin et al. [43] extracted the polarization information of low-resolution PolSAR images and the spatial information of high-resolution single-polarization SAR images. Li et al. [44] designed MCANet to extract high- and low-order independent features using three modules. Thus, complex model structures are often required to obtain better information, which raises challenges related to sensor feature mismatches in SAR and optical image fusion, emphasizing the essential image information without overly complicating the model.
2) *Attention Modules:* Common attention modules include soft attention, hard attention, multihead attention, and self-attention. The soft attention module achieves weighted aggregation of the corresponding output by assigning weights to different parts of the input. Yang et al. [45] introduced a supervised attention module, enabling selective activation of neurons in diagnostic regions while inhibiting irrelevant ones. However, this approach necessitates weighted summarization of all inputs, leading to increased computational complexity. To address this, hard attention module has emerged, focusing on specific input portions at each timestep or step. Wang et al. [46] incorporated a hard attention module into their network to enhance focus on features within "hard" regions of images. However, discrete operations inherent in this approach pose challenges across different subspaces. Multihead attention module has merged, with each head generating attention weights that are combining together. Ho et al. [47] employed a hybrid multihead self-attention (HMS) for final decision-making. Yet, this method may not consider all locations and is influenced by sequence length variations. Self-attention module, which models the dependencies between elements in sequence data, combines with different components to improve the ability of the model to sequence data. Xu et al. [48] developed a triple-attention feature fusion network (TAFFN), capturing spatial and spectral information from SAR and optical images, respectively. Tasnim and Baek [49] extended attention modules to explore intraframe and interframe relations in spatial and temporal dimensions. While self-attention modules offer notable advantages, considering the entire input sequence may lead to model overfitting and semantic ambiguity. Thus, managing model complexity to capture detailed information remains a critical challenge.
3) *Loss Functions:* Various loss functions for image fusion and classification in deep learning can be categorized into several types. The loss function compares all the differences between the true value and the predicted value. Pixel-level loss function directly compares pixel differences. Perceptual loss function utilizes pretrained networks to extract high-level features and compare feature differences. Structural similarity loss function measures the structural similarity. Initially, the loss function typically comprises a single composition, often tailored for a specific problem. Li et al. [44] designed an information entropy loss function that preserves information well for classification tasks. However, as training progresses, the need for more nuanced restrictions becomes apparent, leading to the evolution of diverse loss functions for different phases. Wang et al. [50] applied the tanh function to limit detail loss during encoding, and utilizing pixel-wise loss in the decoding stage to minimize pixel loss. To address computational efficiency while accounting for various concerns, studies on loss function weighting have emerged. Li and Wu [40] introduced the DenseFuse network with a minimization loss function combining pixel and structural loss. Zhao et al. [51] utilized reconstruction and feature decomposition loss to adjust correlation coefficients. Ye et al. [52] proposed a loss function consisting of three parts: pixel loss, structural loss, and texture loss. Different loss functions are investigated to tailor adjustments for respective models, aiming to achieve optimal effects for specific model requirements, making it crucial to study the most suitable and adaptable loss function for proposed models.

Based on the existing methods, although there are quite a few methods for SAR and optical image fusion and intelligent classification based on deep learning, it is evident that certain challenges persist. The following outlines some of the current issues.

Problem 1 (How to Effectively Retain Information During Fusion Due to Mismatched Sensor Characteristics): Existing SAR and optical images have different physical properties and imaging principles, which often leads to a mismatch of sensor features during fusion. Although many studies have attempted to solve this problem by separating higher-order and lower-order features for fusion, this tends to increase the complexity of the model. Therefore, to address the problem of sensor feature mismatch during SAR and optical image fusion, research focuses on preserving the necessary image information without increasing the complexity of the model.

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Problem 2 (How to Obtain Detailed Information of the Obscured Optical Image and Inhibit Loss Caused by Scattering Noise of SAR Image): Optical images are seriously affected by weather, often obscured by clouds or haze, which affects the quality and interpretation ability of the image. SAR images have strong penetration, but serious scattering noise, which will lead to obvious bright and dark patches on the image, making the boundary of the features blurred. When fusion, how to effectively extract this part of the obscured detail information, while effectively reducing the impact of noise becomes a difficult point.

Problem 3 (How to Reduce the Error Caused by the Sample Imbalance for Intelligent Classification): The nonuniform distribution of feature categories may indeed lead to the problem of sample imbalance, which is often encountered in tasks such as remote sensing image classification. Sample imbalance may have a negative impact on model training and performance, so it is necessary to do some processing for samples.

The current study inspires us that in order to solve the above problem, feasible ideas include.

1) *In Terms of Problem 1:* Current researches focus on extracting high- and low-order features of SAR and optical images for cross-fusion through a single encoder. Liang et al. [53] developed the DeFusion to extract high- and low-order features. Zhao et al. [54] proposed the DIDFuse network structure, where the encoder decomposes the image into background and detail feature maps with low- and high-frequency information, respectively. These methods can alleviate the sensor mismatch problem, but good results are often accompanied by increasingly complex network structures. In principle, the dual-encoder structure can obtain the required features of SAR and optical images according to their respective characteristics, which not only can greatly reduce the model complexity, but also effectively solve the problem of sensor mismatch. Therefore, it should be feasible to design a dual-encoder network structure that is suitable for the fusion of SAR and optical images to address the above problem.
2) *In Terms of Problem 2:* Some popular methods include de-clouding [55] and denoising [56], [57], [58], [59], [60], [61], [62] on two images. However, these methods focus on the prefusion period and are difficult to adjust for problems during the fusion process. Another idea is to construct a self-attention module to reconstruct the detailed information of optical images, for example, Fu et al. [63] constructed a dual-attention module to adaptively integrate the local-global relationship. Huang et al. [64] proposed a cross-attention module to obtain the contextual information of all pixels. These methods proved that the attention module can better capture the detailed information, but the effect is not satisfactory when encountering severe occlusion, and it is difficult to take into account the denoising problem of SAR images. Therefore, a feasible solution is to construct a noise-resistant self-attention module, which is put into the SAR encoder to capture the detailed information of the SAR image, suppressing the noise interference while compensating for the problem of occlusion of the optical image.
3) *In Terms of Problem 3:* In previous studies, the loss function was introduced and then reconciled according to practical needs. Chen et al. [65] used a gradient normalized weighted loss function to automatically balance the model training task with improved accuracy. Yeung et al. [66] utilized a new hierarchical framework to construct Unified Focal loss to deal with sample imbalance. These examples verify that the construction of the loss function is very effective for the regulation of network structure and the mitigation of sample imbalance. However, undifferentiated weights often fail to address the loss and sample imbalance issues encountered during fusion and intelligent classification, so an adaptively regulated, combined, and weighted loss function is essential.

Based on the above considerations, we constructed a new fusion network named dual-encoder net (DEN), which consists of two encoders and one decoder. The SAR encoder processes SAR images, utilizing upsampling and downsampling to capture contour information. Simultaneously, the optical encoder handles optical images, extracting texture and detail information through dense convolution operations. Considering the occlusion in optical images and the noise interference in SAR images, we constructed a detail attention module (DAM) against noise interference by taking into account global and local details through parallel convolutional construction of elemental dot products using the intraclass differences between samples caused by SAR noise. It is incorporated into SAR encoder to extract detailed information from the occluded parts of the optical images. A new loss function that adaptively tunes the parameter combinations and weights the categories is constructed to reduce the loss of information and pixels during the fusion process, to stop the interference of image noise, and, most importantly, to reduce the effect of sample imbalance.

The contributions of this article are as follows.

1) *Propose a New Fusion Network DEN:* Compared with other fusion networks together to extract features for SAR and optical images, DEN focuses on extracting the specific features required from each type of image, it effectively simplifies the structure of the fusion network, thereby enhancing fusion speed and improving accuracy simultaneously.
2) *Construct an Attention Module DAM:* DAM not only effectively obtains the detailed information, but also reduces the noise interference. Moreover, it can effectively compensate the occlusion information without increasing the complexity of the model for the fusion of SAR and optical images.
3) *Propose a New Loss Function:* Comparing with the existing popular loss functions, this loss function is very effective in preserving the image information, improving the quality of fused images, and classification accuracy.

The remainder of this study is structured as follows. Section II outlines the related study. Section III presents and analyzes the experimental findings. Finally, the conclusion of this study is provided in the last section.

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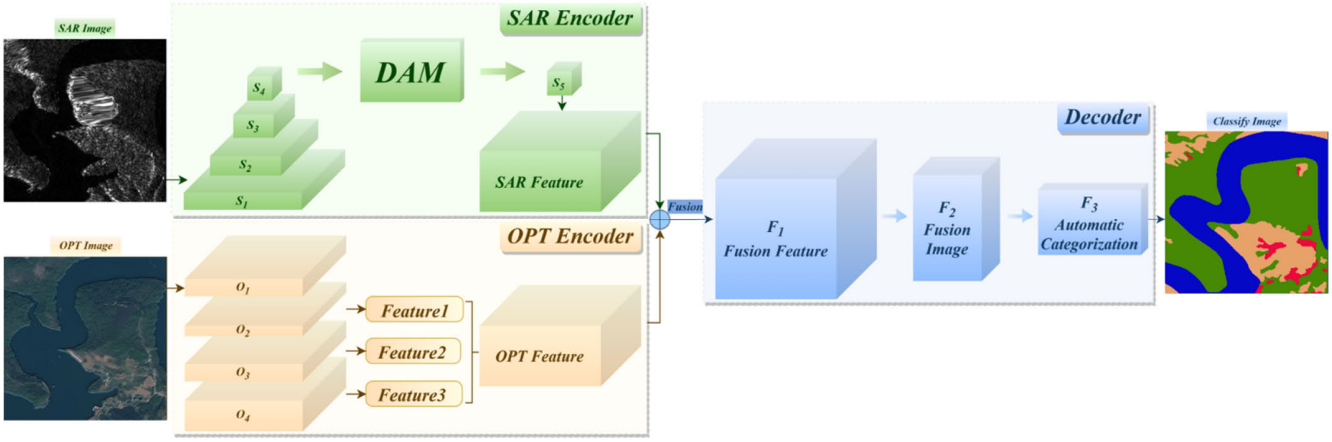


Fig. 1. DEN infrastructure. SAR Image denotes the input SAR image, S_1-S_5 denote the process of SAR image transformation, DAM denotes the detail attention module, and SAR Feature denotes the SAR features extracted by the SAR encoder. OPT Image denotes the input optical image, O_1-O_4 denote the process of optical image transformation, Features 1–3 denote the detail features extracted layer by layer from the optical image, and OPT Feature denotes the optical features extracted by the optical encoder. F_1-F_3 denote the process of decoding and transforming the fused image, and the Classify Image denotes the final classified result.

II. Proposed Methodology

In this section, the designed network will be described in detail, which contains three main parts: the overall fusion network structure, the attention module, and the loss function.

A. Dual-Encoder Net

The proposed fusion network structure DEN consists of two parts, encoder and decoder, as shown in Fig. 1. The left side represents the encoder, while the right side depicts the decoder. The encoder consists of two parts: the SAR encoder and the optical encoder, arranged from top to bottom.

SAR images and optical images possess distinct characteristics, offering complementary insights into the image region and enabling the generation of images with diverse spatial and spectral information. Previous research commonly involved feeding SAR and optical images into the same encoder to extract their higher-order and lower-order features, respectively, often overlooking their unique traits. Hence, this study designs separate SAR and optical encoders to individually extract their distinctive features, which are subsequently inputted into the same decoder for decoding.

SAR images are characterized by excellent semantic information, high resolution, strong penetration, and clear contours. Consequently, an SAR autoencoder is devised, with an initial input SAR encoder size of 320 × 320 × 32. Initially, downsampling is employed, reducing the image dimensions to 160, 80, and 40, respectively, while adjusting the channel numbers to 32, 64, 128, and 32, finally upsampling to yield a 320 × 320 × 64 output encoder. This process facilitates semantic feature extraction through continuous upsampling and downsampling, effectively altering the image dimensions. In contrast, optical images boast rich texture and detail information, with texture features continually extracted via convolution and dense linking operations.

Optical images are rich in texture and detail information, and features can be continuously extracted by convolution and dense linking operations. The input size of the optical autoencoder is 320 × 320 × 32, and the base features of 320 × 320 × 16 size are first convolved to obtain the

Algorithm 1 DEN

Input: E: Training epochs; B: batch size; S: input SAR image; O: input OPT image; τ: sample batch; ξ: learning rate; α, β: trade-off hyperparameter

Output: model parameter η

randomly initialize η;
calculate class weights;
repeat
  for each task $\tau_i \in \tau$ **do**
    sample $\{S, O\} \sim \tau_i$
    SAR Encoder: calculate input SAR image S $\rightarrow$ feature maps: S_1, S_2, S_3, S_4;
    DAM: get $Feature_{SAR}$ using $S_4 \rightarrow$ get $Attention_{SAR}$ using parallel convolution D_1, $D_2 \rightarrow$ Details in SAR image: $Detail_{SAR}$;
    SAR Feature: obtain feature map: $S_5 \rightarrow$ Upsample to obtain SAR feature;
    OPT Encoder: calculate input SAR image $O \rightarrow$ feature maps: O_1, $O_2 \rightarrow$ calculate Feature1 $\rightarrow$ feature map: $O_3 \rightarrow$ calculate Feature2 $\rightarrow$ feature map: $O_4 \rightarrow$ calculate Feature3 $\rightarrow$ obtain OPT feature;
    Decoder: cat SAR feature and OPT feature to obtain Fusion Feature $\xrightarrow{Conv,BatchNorm,ReLU}$ Fusion image $\xrightarrow{ConvTranspose}$ Automatic Categorization;
    $L_{\text{total}} = L_I + \alpha L_P + \beta L_N$;
    Update model: $\eta \leftarrow \eta - \xi * \nabla_\eta L$;
  end for;
until convergence;

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TABLE I
DEN PARAMETERS

Block		Layer	Shape (input)	Shape (output)	Channel (input)	Channel (output)
Encoder	SAR Encoder	S_1	320	320	1	32
		S_2	320	160	32	64
		S_3	160	80	64	128
		S_4	80	40	128	32
		DAM	40	40	32	32
		S_5	40	40	32	64
		SAR Feature	40	320	64	64
	OPT Encoder	O_1	320	320	3	32
		O_2	320	320	32	16
		Feature1	320	320	16	16
		O_3	320	320	16	32
		Feature2	320	320	32	16
		O_4	320	320	16	48
		Feature3	320	320	48	16
		OPT Feature	320	320	16	64
Decoder		F_1	320	320	64	128
		F_2	320	320	128	64
		F_3	320	320	64	6

16-channel Feature1 by one convolution and normalized Mish activation function, and then the base features and Feature1 are fused to obtain the 32-channel feature layer, and then the 16-channel Feature2 is obtained by the same operation of convolution and normalized Mish activation function. Then the base features, Features 1 and 2 are fused to obtain a 48-channel feature layer, and the 16-channel Feature3 is obtained by the same convolutional, normalized Mish activation function operation. Finally, the base features, Features 1–3 are fused to obtain a 64-channel optical feature output encoder. This output contains the optical texture details and spectral features of the image.

The feature images outputted from the respective encoders are fused and fed into the decoder, generating a Fusion Feature sized $320 \times 320 \times 128$. Following convolution, ReLU activation, and normalization, the Fusion Feature forms the Fusion Image. Subsequently, ConvTranspose2d deconvolution processing and decoding are applied to the Fusion Image to derive the outcome of Automatic Categorization. Table I displays the network parameters.

B. Detail Attention Module

After fusion, the texture details in some areas were not clear enough, which prompted us to propose DAM. Considering that optical images already undergo thorough feature extraction, other unclear texture details likely originate from the SAR image. Leveraging its robust penetration capability, SAR imagery can effectively capture feature information even amidst cloud cover or occlusion. Therefore, DAM tailored for SAR imagery is devised and incorporated into the SAR encoder, as depicted in Fig. 2, outlining the specific structure of the proposed DAM.

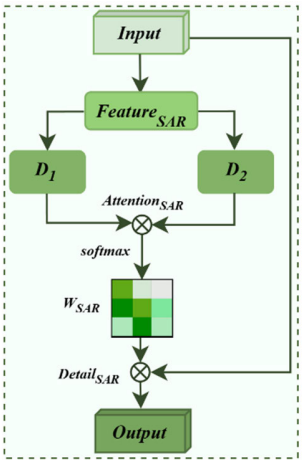


Fig. 2. DAM. Feature$_{SAR}$ denotes the given SAR features, D_1, D_2 denote the two spatial queries generated, Attention$_{SAR}$ denotes the feature attention map of the SAR image, W_{SAR} denotes the weights of each feature, and Detail$_{SAR}$ denotes the detail feature attention map of the SAR image.

In SAR image, the interference from scattering noise is particularly pronounced, resulting in diverse manifestations within the same class of samples. To address this challenge, DAM is devised to encode remote information into domain features by discerning contextual relationships. This approach aims to enhance the recognition of intraclass relationships, accentuate spatial features within the class, and suppress noisy features, thereby mitigating the impact of discrepancies. In the module, given a feature $\text{Feature}_{\text{SAR}} = \{f_{\text{SAR}}\}_{i=1}^{n} \in R^{B\times C\times H\times W}$, where B denotes the batch size, C denotes the number of channels, H and W denote the height and width, respectively, and i denotes the image number. First, two spatial queries $\{D_1^i, D_2^i\}_{i=1}^{n} \in R^{B\times C'\times H\times W}$ are generated synchronously using two parallel

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extended convolutions, C' denotes the number of channels after passing the convolution, which can not only increase the local receptive field without increasing the parameters, but also control the running speed without increasing the complexity of the model. In this process, D_1 reconstruct and transpose into $\{D_1^i\}_{i=1}^n \in R^{B\times Q\times C'}$, where $Q = H \times W$, and then D_2 does the same reconstruction and then dot-multiply, to obtain the feature attention map $\text{Attention}_{\text{SAR}}$, $\{\text{Attention}_{\text{SAR}}^i\}_{i=1}^n \in R^{B\times Q\times Q}$ of an SAR image, softmax operation is applied to the feature attention map to obtain the weights W_{SAR} of each feature, representing the correlation between the features

$$W_{\text{SAR}}^i = \frac{\exp\left[\left(D_1^i\right)^{\text{T}} \times D_2^i\right]}{\sum_{j=1}^{Q} \exp\left[\left(D_1^{i,j}\right)^{\text{T}} \times D_2^{i,j}\right]} \tag{1}$$

where $D_1^{i,j}$ and $D_2^{i,j}$ denote the two spatial queries of the jth pixel of the ith image, and T denotes the transpose operation. Next, $\text{Feature}_{\text{SAR}}$ is reconstructed into $R^{B\times C\times Q}$, and the result is element-wise multiplied with the transpose of W_{SAR}. Finally, the multiplication result is reconstructed and embedded to obtain the feature attention map $\text{Detail}_{\text{SAR}}$ for SAR image details

$$\text{Detail}_{\text{SAR}}^i = \gamma \sum_{j=1}^{Q} f_{\text{SAR}}^{i,j} \cdot \left(W_{\text{SAR}}^{i,j}\right)^{\text{T}} \tag{2}$$

where the feature attention map $\text{Detail}_{\text{SAR}}$ for SAR image details contains weighted sums of all spatial features, which enables better recognition of intraclass relationships and emphasizes intraclass spatial features. γ is a self-learning parameter that can self-adjust through learning.

C. Loss Function

A new loss function is designed, comprising three components: information loss (L_{I}), pixel loss (L_{P}), and noise loss (L_{N}), as shown in the following equation:

$$L_{\text{total}} = L_{\text{I}} + \alpha L_{\text{P}} + \beta L_{\text{N}} \tag{3}$$

where L_{I} represents the information loss, L_{P} denotes pixel loss, L_{N} refers to noise loss. There are two tuning parameters, α and β, which are ultimately selected as 1 and 5 depending on the adaptive tuning.

The information loss refers to the information entropy between the fused image and the source image. The magnitude of information is inversely proportional to the probability of occurrence. If the probability of an event is $A(x)$, then the information quantity $I(x)$ is represented as

$$I(x) = -\log(A(x)). \tag{4}$$

The expected information entropy $H(x)$ of all information quantities can be represented as

$$H(X) = -\sum_{i=1}^{n} P(x_i)\log(P(x_i)), \quad (X = x_1, x_2 \ldots, x_n). \tag{5}$$

If the same random variable has two independent probability distributions $P(x)$ and $Q(x)$, where $p(x_i)$ represents the ith sample true distribution and $q(x_i)$ represents the model predictive distribution for the ith sample, the difference between the two distributions can be represented by the relative entropy D_{KL}, also known as the Kullback–Leibler (KL) [67] divergence, defined as

$$D_{\text{KL}}(p||q) = \sum_{i=1}^{n} p(x_i)\log\left(\frac{p(x_i)}{q(x_i)}\right). \tag{6}$$

The formula for relative entropy can be broken down as follows:

$$\begin{aligned} D_{\text{KL}}(p||q) &= \sum_{i=1}^{n} p(x_i)\log\left(\frac{p(x_i)}{q(x_i)}\right) \\ &= \sum_{i=1}^{n} p(x_i)\log(p(x_i)) - \sum_{i=1}^{n} p(x_i)\log(q(x_i)) \\ &= -H(p(x)) + \left[-\sum_{i=1}^{n} p(x_i)\log(q(x_i))\right]. \end{aligned} \tag{7}$$

The former $H(x)$ is the entropy, and the latter is the cross-entropy, which is equal to the sum of the relative entropy and the entropy, expressed as

$$H(p,q) = -\sum_{i=1}^{n} p(x_i)\log(q(x_i)). \tag{8}$$

Cross-entropy quantifies the difference between two probability distributions of the same random variable, typically reflecting the variance between the true and predicted probability distributions in machine learning. In deep learning networks, entropy remains constant, and lower relative entropy usually signifies better predictive performance. However, cross-entropy computation is comparatively simpler, making it a commonly adopted loss function for classification tasks. In the proposed method, we calculate the information cross-entropy between the weighted classification outcomes of the fused image and the ground truth. Through the utilization of cross-entropy loss, we iteratively refine and optimize the information within the fused image, effectively preserving its informational richness.

The pixel loss L_P is computed by first determining the maximum pixel values of SAR and optical images, and then calculating the L_1 norm loss between the target and model output values

$$L_{\text{P}} = \frac{1}{HW}\left\|I_f - \max(I_{\text{SAR}}, I_{\text{OPT}})\right\|_1 \tag{9}$$

where I_f denotes the fusion feature, I_{SAR} denotes the SAR feature output from the SAR encoder, and I_{OPT} denotes the optical feature output from the optical encoder.

The noise loss L_N

$$L_{\text{N}} = \frac{1}{HW}\left\|\left|\nabla I_f\right| - \max(|\nabla I_{\text{SAR}}|, |\nabla I_{\text{OPT}}|)\right\|_1 \tag{10}$$

where ∇ represents the Sobel gradient operator, which works well for images with significant grayscale gradient noise and provides accurate edge positioning. Both the target value and the model output value are convolved with the Sobel operator before computing the L_1 norm loss.

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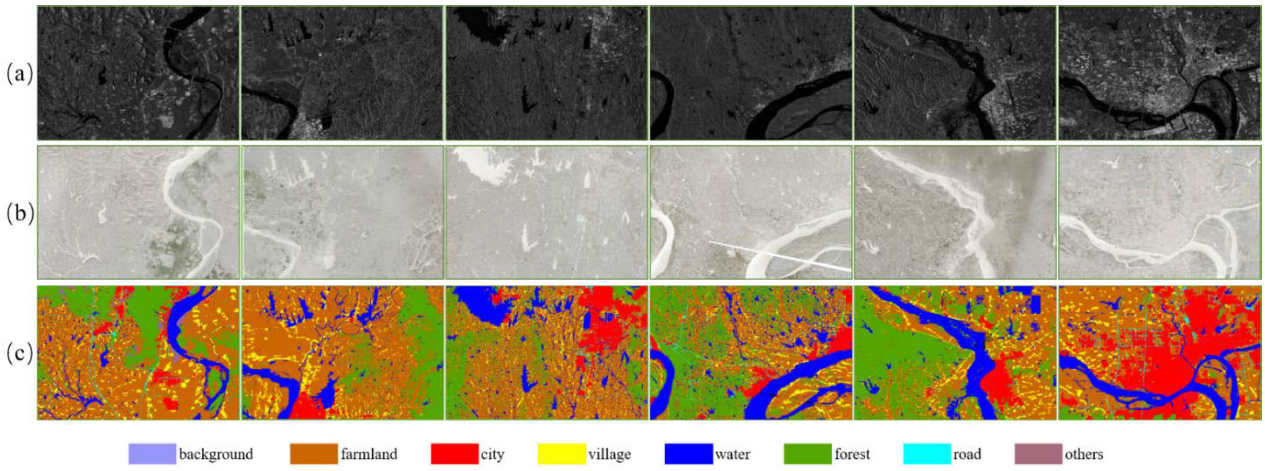


Fig. 3. Partial images of the original dataset. (a) SAR images. (b) Optical images. (c) Labeled images.

In addition, to address the issue of sample imbalance, statistical calculations are employed to determine the sample sizes of each category. Subsequently, the percentage of samples within each category is utilized to assign weights, wherein categories with a higher number of samples are assigned lower weights, while those with fewer samples are assigned higher weights. These statistically derived weight values are then incorporated into the loss function, effectively mitigating the problem of sample imbalance.

D. McNemar Test

The McNemar test [68] is a statistical test for comparing the difference in performance between two classifiers or models on a dichotomous classification task. The classification results of the two models on the same samples are counted by constructing a 2 × 2 list of columns as shown in Table II. The specific weight assignments are detailed in Table III.

Calculate the test statistic mainly related to *b, c*

$$\chi^2_{Mc} = \frac{[|(b-c)|-1]^2}{b+c} \tag{11}$$

where b denotes the number of samples correctly classified by model 1 but incorrectly classified by model 2, and c denotes the number of samples incorrectly classified by model 1 but correctly classified by model 2, χ^2_{Mc} statistic approximately follows a χ^2 distribution with 1° of freedom. The p-value of the chi-square distribution is found using the calculated statistic. Typically, for a chi-square distribution with 1° of freedom, the critical value for a significance level of 0.05 is 3.841 and the critical value for a significance level of 0.01 is approximately 6.635.

TABLE II
2 × 2 COLUMNAR TABLE CONSTRUCTED BY THE MCNEMAR TEST

	Model 2 Positive	*Model 2* Negative
Model 1 Positive	*a*	*b*
Model 1 Negative	*c*	*d*

TABLE III
NUMBER OF PIXELS AND PENALTY WEIGHTS FOR EACH CATEGORY

Number	Category	Pixels Number	Weight
0	forest	431486685	0.0252
1	field	396141451	0.0275
2	water	163945591	0.0664
3	building	120265056	0.0904
4	road	10878261	1.0000
5	others	26210956	0.4150

III. EXPERIMENTS

A. Datasets and Experimental Settings

The dataset utilized in the experiment is the WHU-OPT-SAR open-source dataset, specifically curated for SAR and optical image fusion tasks. This dataset comprises 100 pairs of SAR and optical images, each with dimensions of 5556 × 3704 pixels, depicting the same geographical region. The region covered spans approximately 50 000 km^2 within Hubei, China, ranging from latitudes 30°N–33°N. This area boasts a favorable climate and exhibits minimal variation in elevation, ranging from 50 to 3000 m. Moreover, it encompasses diverse terrains, rendering it well-suited for conducting fusion-related studies.

The original image measures 5556 × 3704 in size and is segmented into eight distinct categories: background, farmland, city, village, water, forest, road, and others, as illustrated in Fig. 3.

First, image cropping was performed on all SAR and optical images, as well as the labeling images. From the original dataset of 100 images, each sized 5556 × 3704 pixels, a total of 18 700 images were cropped to a size of 320 × 320 pixels. Subsequently, the dataset was divided into training, validation, and test sets in a 6:2:2 ratio, with each set comprising 11 220, 3740, and 3740 images, respectively. The image categories were then reclassified into six categories: forest, field, water, building, road, and others. The processed image data is depicted in Fig. 4, which was utilized for subsequent experiments.

Finally, to solve the sample imbalance problem, as shown in Table III. The total number of pixels in each category was

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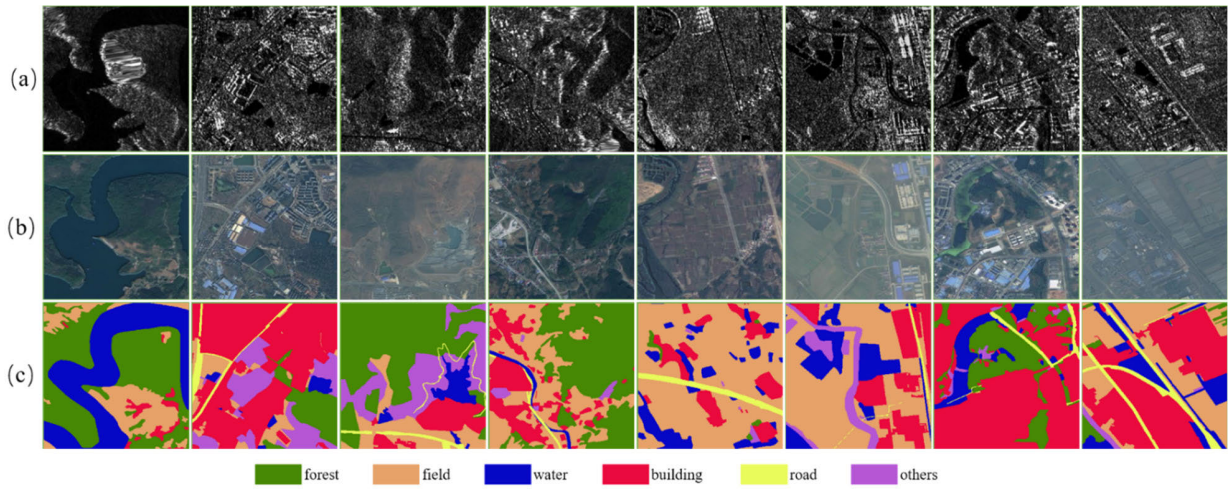


Fig. 4. Dataset presentation of different images. (a) SAR images. (b) Optical images. (c) Labeled images.

first counted. Then the percentage of each category in the total pixels is calculated. The inverse of the proportion is then used as the weight of each category, and the smaller the proportion of pixels in a category, the larger the weight. Finally, the category weights are normalized using the maximum value to get the final penalty weights. The sample categories with less total pixels are made to have higher weights to get more attention.

All experiments in this article use PyTorch framework and NVIDIA GeForce RTX3090Ti processor, set batch size to 8, weight decay set to 0.0001, initial learning rate to 0.001 Adam algorithm to optimize the loss function, and total epochs set to 50, there is no pretraining, and the loss weights are the values in the penalty weight column of Table III.

B. Evaluation Criteria

To comprehensively evaluate the fusion effect, various evaluation metrics are employed in this study. These include overall accuracy (OA), which represents the proportion of correctly classified samples out of the total; intersection over union (IoU), measuring the overlap between predicted and true regions; KAPPA coefficient, assessing classification consistency; floating point operations per second (Flops) refers to the number of floating-point operations a model can perform per second, typically used to measure the computational complexity of a model; Model parameters (Params) refers to the number of trainable weights and biases in the model, and is mainly used as a measure of the capacity and complexity of the model; entropy (EN), indicating image information content; standard deviation (SD), reflecting pixel distribution around the mean; spatial frequency (SF), representing grayscale change rate; peak signal-to-noise ratio (PSNR), quantifying image noise; and mean square error (MSE), sensitive to fusion outliers and measuring fusion quality.

C. Comparison of Classification Performance

The comparative experiments are divided into three parts, each validating the proposed fusion and intelligent classification model, the attention module, and the loss function, respectively. Across all three sets of control experiments, the proposed method consistently demonstrates enhanced fusion detection and classification performance compared to existing popular methods.

1) Dual-Encoder Net: In this section, comparative experiments were conducted using several existing popular image fusion networks on the WHU-OPT-SAR dataset. These include the simple network structure DenseFuse [40], as well as the complex network structures DeFusion [53], DIDFuse [54], include the simple network structure DenseFuse [40], as well as the complex network structures DeFusion [53], DIDFuse [54], and PSFN [43], which utilize the extraction of high- and low-order features for fusion. In addition, the SAR and optical image fusion network structure MCANet [44] was included for comparison. A total of five networks were compared with the proposed DEN, all using the same dataset and loss function, with only the network structure varying. The results of intelligent classification are illustrated in Fig. 5.

The results highlight the favorable performance of the proposed network structure for image classification. Specifically, in Fig. 5(d), the simple network DenseFuse exhibits some classification confusion in most labeled areas, presenting a "false clear" state despite its simplicity. Fig. 5(e)–(g) depicts the classification outcomes of the complex networks DeFusion, DIDFuse, and PSFN, respectively. These networks, which extract high- and low-order features from the fused images, show various issues: DeFusion features numerous classification omissions, DIDFuse misclassifies a significant number of building categories, and although PSFN boasts the highest model complexity, its classification effectiveness is indeed improved, yet, even when comparing with the DEN, there remain some small-scale recognition errors. In Fig. 5(h), the land classification network structure MCANet exhibits instances of small-scale misrecognition and some large-scale classification omissions. These issues stem from insufficient feature information, often leading to classification failures and misallocating the original region categories into others.

For the labeled areas in Fig. 5, it is evident that the proposed network structure closely aligns with the actual labeled categories, while all other network structures exhibit some form of omission or misclassification. In the first row, the DenseFuse network incorrectly identifies the actual field area as building

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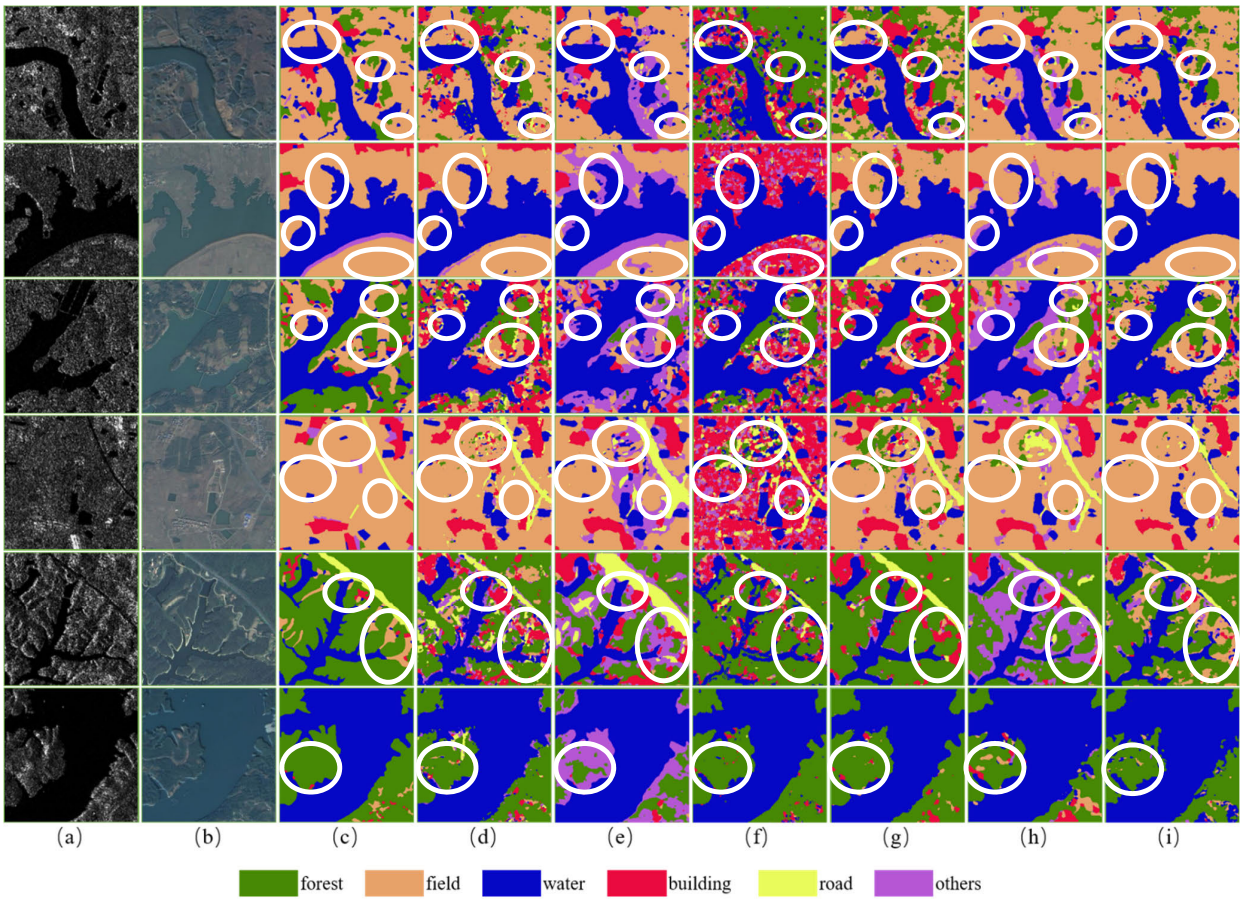


Fig. 5. Image fusion detection results for each network structure. (a) SAR images. (b) Optical images. (c) Labeled images. (d) DenseFuse classification results. (e) DeFusion classification results. (f) DIDFuse classification results. (g) PSFN classification results. (h) MCANet classification results. (i) DEN (ours) classification results.

and road categories. In the second row, the DeFusion network overlooks the classification of the large-scale field area and fails to recognize the small-scale water category. Moving to the third row, the left-labeled area, originally a field, is misclassified by DIDFuse as water, with additional building and road categories. Furthermore, DIDFuse incorrectly categorizes the top-labeled field and forest regions as water, building, and road categories, along with an erroneous forest classification. Similarly, the bottom-labeled region is inaccurately classified as an impractical mix of small categories. In the fourth row, representing actual field and water areas, PSFN incorrectly identifies a large forest category and small portions of building and road categories. In the fifth row, MCANet fails to correctly classify the forest and road categories, while in the sixth row, where the actual forest category is present, MCANet identifies field, water, and building categories. Therefore, based on qualitative analysis, the proposed network structure for fusion classification demonstrates superior results.

Table IV shows the results of fusion detection classification of six network structures, and it can be seen that the proposed network structure has the highest OA, which is 1.4% higher than DenseFuse, 2.4% higher than DeFusion, 53.6% higher than DIDFuse, 8.6% higher than PSFN, and 2.3% higher than MCANet. Meanwhile, the proposed network structure has the lowest number of Flops, the highest computational efficiency, the fastest inference speed, 15.3% faster than DenseFuse, 54.2% faster than DeFusion, 79.5% faster than DIDFuse, 96.2% faster than PSFN, and 90.0% faster than MCANet. The proposed method models the least number of Params, which is 7.1% less than DenseFuse, 96.8% less than DeFusion, 3.7% less than DIDFuse, 91.3% less than PSFN, and 99.5% less than MCANet. In other words, the proposed network structure achieves higher detection accuracy without increasing the model complexity compared to other network structures.

TABLE IV
TEST RESULTS FOR SIX NETWORK STRUCTURES

Model	OA	IOU	KAPPA	Flops (M)	Params (M)
Densefuse [40]	0.763	0.332	0.539	105093	0.28
DeFusion [53]	0.753	0.357	**0.549**	194268	8.06
DIDFuse [54]	0.241	0.144	0.136	433291	0.27
PSFN [43]	0.691	0.337	0.473	2369214	2.99
MCANet [44]	0.754	**0.368**	0.527	890904	55.79
DEN	**0.777**	0.300	0.509	**89001**	**0.26**

Table V shows the results of different evaluation indexes after fusion classification of the six network structures, and it can be seen that the proposed method achieves the highest values in EN and SD, and the lowest value in MSE, which all prove that the fusion effect is the best. The EN is 0.21 higher than DenseFuse, 0.25 higher than DeFusion,

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TABLE V
RESULTS OF DIFFERENT EVALUATION INDICATORS FOR THE SIX NETWORK STRUCTURES

Model	EN	SD	SF	PSNR	MSE
Densefuse [40]	11.22	32.77	21.93	5.94	1.44
DeFusion [53]	11.18	32.23	16.01	6.97	1.56
DIDFuse [54]	11.27	31.52	**27.90**	**7.29**	2.89
PSFN [43]	11.16	30.70	19.09	5.79	2.61
MCANet [44]	11.33	28.44	14.68	6.02	1.58
DEN	**11.43**	**34.02**	21.94	6.03	**1.43**

0.16 higher than DIDFuse, 0.27 higher than PSFN, and 0.10 higher than MCANet, which indicates that the proposed network structure fuses the most informative images. The SD is 1.25 higher than DenseFuse, 1.79 higher than DeFusion, and 2.50 higher than DIDFuse, 3.32 higher than PSFN, and 5.58 higher than MCANet, indicating that the proposed network structure fusion has the highest image quality and the clearest image. The MSE is 0.01 lower than DenseFuse, 0.13 lower than DeFusion, 1.46 lower than DIDFuse, 1.18 lower than PSFN and 0.15 lower than MCANet, the proposed method fuses out the least number of image outliers. The SF and PSNR of DIDFuse are higher than that of the proposed method, and only from the point of view of these two values, it is better than the proposed method, but in combination with Fig. 5 and the OA, it can be seen that it presents "pseudo-clearness," with classification errors in a large area. In addition to this method, the proposed method has the highest SF, which is 0.01 higher than DenseFuse, 5.93 higher than DeFusion, 1.32 higher than DIDFuse, 2.85 higher than PSFN, and 7.26 higher than MCANet, which indicates that the proposed method fuses images with high gray level change rate and the best image quality. DeFusion network is slightly higher than the proposed method in PSNR, indicating that the proposed method fuses images with insufficient processing of noise effects, and the subsequent proposed noise loss does a good job of dealing with this problem. Taken together, the fused image of the proposed method performs optimally in terms of fusion quality compared to the current state-of-the-art fusion methods.

Table VI shows the evaluation results using McNemar's test of DEN and five network structures. Because the size of the data volume has a significant impact on the significance test, we completed the evaluation by controlling the number of samples through both random sampling and stratified sampling. First, we randomly sampled by determining the value of b as 100, 500, 1000, and 5000 to get the statistics and then determine the corresponding p-value, at significance 0.01, our model is basically significantly different from each model and $b > c$, i.e., DEN significantly outperforms the other models. However, $b = 100$ appeared no significant difference between DEN and MCANet ($p = 0.0696$) classification results, and at $b = 500$, there was no significant difference between DEN and DeFusion (0.4046), DEN and DIDFuse (0.0543) classification results, which was analyzed to be due to the random sampling, the number of samples is too small in the overall number, the stability is poor and data fluctuation is large, but the undeniable point is that b is always greater than c, which shows that the classification effect of DEN is always better than other models. The latter set of experiments we used stratified sampling, respectively, taking one layer and four layers of undifferentiated experiments, the final results obtained, regardless of one layer or four layers, significance of 0.01 case, DEN is significantly better than other models.

Table VII shows the detection accuracies of the six network structures in different categories, and Table VIII shows the KAPPA coefficients of the six network structures in different categories, the proposed network structures achieved higher values in different categories, respectively, both proving that the proposed network structures are better in classification. In the forest category, the OA value of classification detection accuracy is 8.46% higher than DenseFuse, 18.59% higher than DeFusion, 8.06% higher than DIDFuse, 1.77% higher than PSFN, and 13.72% higher than MCANet. In the field category, the OA value of classification detection accuracy is 6.39% higher than DenseFuse, 11.80% higher than DeFusion, 84.10% higher than DIDFuse, 19.40% higher than PSFN, and 5.89% higher than MCANet. The KAPPA coefficient of others category, which is 2.05 times higher than that of DenseFuse, 4.73 times higher than that of DeFusion, 2.16 times higher than that of DIDFuse and PSFN, 1.98 times higher than that of MCANet. The remaining categories of water, building, and road, although not the highest, have about the same accuracy as the other network classifications, and it can be seen that as the number of categories decreases, especially road has the least number of categories, the accuracy of the classification is comparatively worse, therefore, the analysis is that the results are unsatisfactory due to the sample imbalance, so that after the subsequent weighted processing of the loss function, the problem was better solved. Overall, from the perspective of quantitative analysis, the proposed network structure has a great improvement in detection and classification compared with the other five state-of-the-art network structures.

2) Detail Attention Module: In this section, the experiments are still completed on the WHU-OPT-SAR dataset, and it can be seen that the detailed information of some regions in the proposed network structure is not well recognized, which is analyzed to be due to the missing details caused by the optical image being interfered with by the occlusion or the weather, whereas the SAR image has a very strong penetrating ability, which is not affected by the time, weather, and occlusion. Therefore, DAM is proposed to be added to the SAR encoder.

From the results in Fig. 6 below, it can be seen that (d) is the result without adding any DAM, (e) is the result of adding DAM to both SAR and optical images, (f) is the result of adding DAM to optical images only, and (g) is the result of adding DAM to SAR images only. When No-DAM, as in Fig. 6(d), many regions are classified more chaotically, and a lot of detailed information is not well recognized. In the first row, it can be seen that the upper left corner is actually of field, but it is not recognized, which is mixed with forest next to it. In the middle of water around the area which should be forest, are all judged as background and not successfully recognized. The field in the lower right corner, again a large part of the area recognized as background and other classes, and a part

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TABLE VI
EVALUATION RESULTS USING MCNEMAR'S TEST OF DEN AND FIVE NETWORK STRUCTURES

Model	$b = 100$		$b = 500$		$b = 1000$		$b = 5000$	
	χ^2_{Mc}	p	χ^2_{Mc}	p	χ^2_{Mc}	p	χ^2_{Mc}	p
Densefuse [40]	98.01	$p < 0.0001$	498.00	$p < 0.0001$	998.00	$p < 0.0001$	3742.86	$p < 0.0001$
DeFusion [53]	25.89	$p < 0.0001$	0.69	$p \approx 0.4046$	15.98	$p < 0.0001$	276.80	$p < 0.0001$
DIDFuse [54]	28.06	$p < 0.0001$	3.70	$p \approx 0.0543$	72.42	$p < 0.0001$	1546.01	$p < 0.0001$
PSFN [43]	12.49	$p \approx 0.0004$	11.84	$p \approx 0.0005$	115.99	$p < 0.0001$	1046.86	$p < 0.0001$
MCANet [44]	3.29	$p \approx 0.0696$	63.43	$p < 0.0001$	90.43	$p < 0.0001$	155.33	$p < 0.0001$
Model	**Layer = 1**				**Layer = 4**			
	b	c	χ^2_{Mc}	p	b	c	χ^2_{Mc}	p
Densefuse [40]	24713	5384	12412.25	$p < 0.0001$	112405	12575	79739.39	$p < 0.0001$
DeFusion [53]	6759	4499	1453.28	$p < 0.0001$	12298	11960	4.68	$p \approx 0.0305$
DIDFuse [54]	9706	4218	2162.25	$p < 0.0001$	13014	11589	82.42	$p < 0.0001$
PSFN [43]	8590	3714	1931.54	$p < 0.0001$	12766	10677	185.97	$p < 0.0001$
MCANet [44]	6021	4720	157.34	$p < 0.0001$	14480	9692	948.01	$p < 0.0001$

TABLE VII
DETECTION ACCURACIES FOR DIFFERENT CLASSES OF THE SIX NETWORK STRUCTURES

OA	Forest	Field	Water	Building	Road	Others
Densefuse [40]	0.1301	0.8431	0.6346	0.6480	0.5627	0.0017
DeFusion [53]	0.0288	0.7890	**0.7286**	0.6952	**0.9048**	0.0211
DIDFuse [54]	0.1341	0.0660	0.7265	0.6801	0.4684	**0.1931**
PSFN [43]	0.1970	0.7130	0.6428	**0.7654**	0.6663	0.0380
MCANet [44]	0.0775	0.8481	0.5433	0.5875	0.7941	0.1110
DEN	**0.2147**	**0.9070**	0.6631	0.4126	0.1984	0.0549

TABLE VIII
KAPPA COEFFICIENTS FOR DIFFERENT CATEGORIES OF THE SIX NETWORK STRUCTURES

KAPPA	Forest	Field	Water	Building	Road	Others
Densefuse [40]	**0.0612**	0.7672	0.4916	0.4321	0.2399	0.0006
DeFusion [53]	0.0263	0.7415	0.4717	**0.4816**	0.4224	0.0026
DIDFuse [54]	0.0435	0.0650	0.3767	0.1455	0.2304	0.0057
PSFN [43]	0.0380	0.6701	**0.5135**	0.4060	0.3901	0.0057
MCANet [44]	0.0511	0.7753	0.4708	0.4554	**0.4535**	0.0062
DEN	0.0119	**0.7799**	0.4651	0.4591	0.1715	**0.0123**

of it is mixed with the convenient building. When Both-DAM, it instead results in a classification error, as shown in Fig. 6(e), in the second row, it can be seen that the area is actually the field is recognized as a large area of forest. In the third row, the area that is actually the field is recognized as forest and building. When OPT-DAM, as in Fig. 6(f), in the fourth row, the region that is actually field in the left and upper right corners is categorized into the background class and the forest class; in the middle, the region that is actually forest is not successfully recognized and is categorized into the background class; in the lower right corner that is actually forest is partially categorized into the background and partially categorized into the building class next to it. In the fifth row that is actually the forest class, which is classified into the background. When SAR-DAM, as in Fig. 6(g) the correct class can be recognized well. In the sixth row, it can be seen that when No-DAM, some details fail to be recognized. When Both-DAM, the effect is worse, resulting in many misclassifications. When OPT-DAM, the correct categories are not well recognized. Finally, SAR-DAM can recognize more details and reduce misclassifications, which also verifies the conjecture that the missing detail information in the proposed network structure comes from the SAR image, and SAR-DAM can better extract the detail features of the SAR image, which can make the fusion classification effect better.

Table IX shows the results of adding the DAM test, and it can be seen that SAR-DAM has the highest detection accuracy. The SAR-DAM improves the average detection

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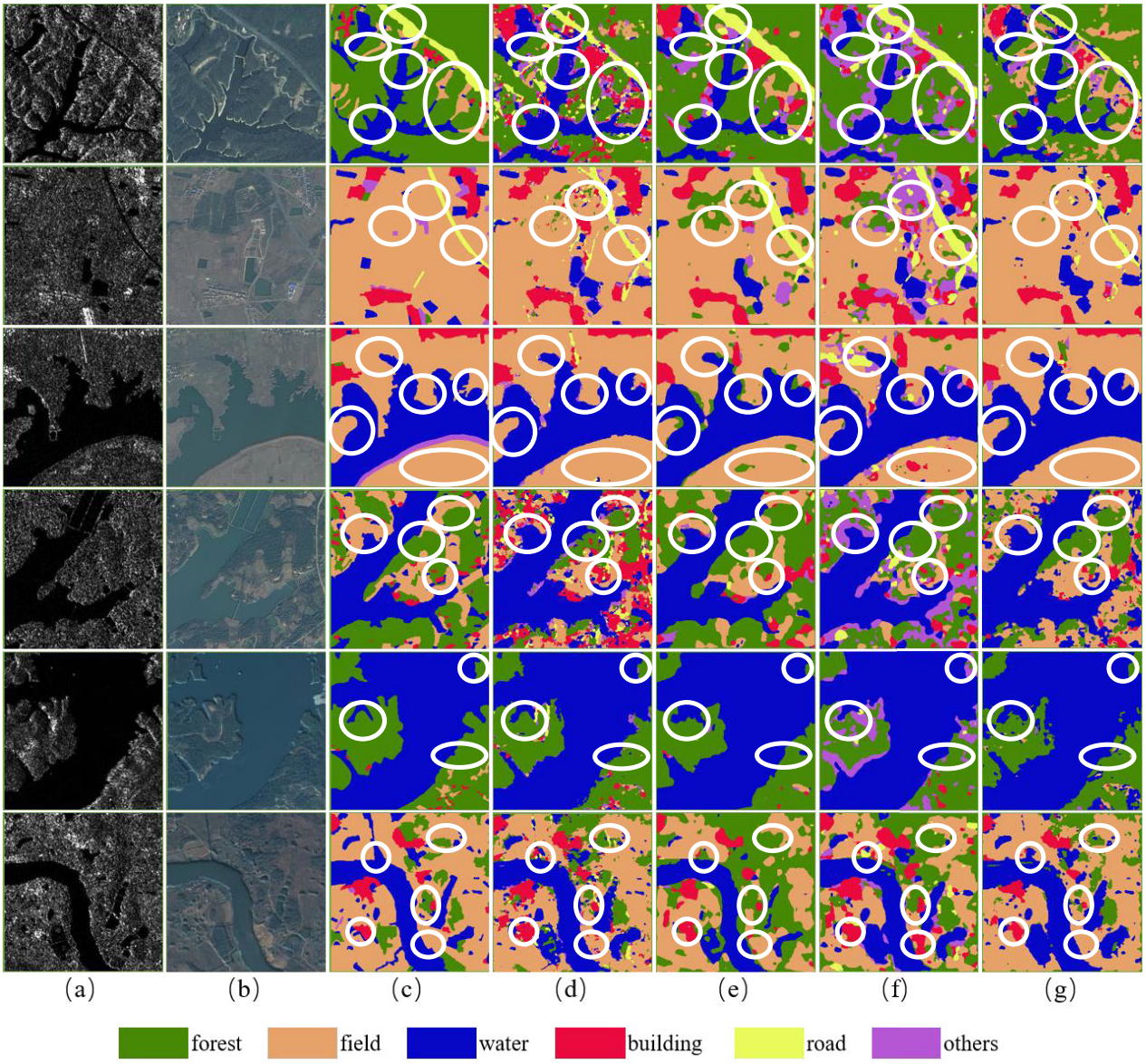


Fig. 6. Classification results of fusion detection by adding attention module. (a) SAR images. (b) Optical images. (c) Labeled images. (d) No-DAM classification results. (e) Both-DAM classification results. (f) OPT-DAM classification results. (g) SAR-DAM (ours) classification results.

TABLE IX
ADDITION OF ATTENTION MODULE TEST RESULTS

Attention	OA	IOU	KAPPA
No - DAM	0.759	**0.336**	**0.535**
Both - DAM	0.575	0.286	0.319
OPT - DAM	0.627	0.296	0.398
SAR - DAM	**0.777**	0.300	0.509

TABLE X
RESULTS OF DIFFERENT EVALUATION INDICATORS FOR ACCESSION TO THE ATTENTION MODULES

Attention	EN	SD	SF	PSNR	MSE
No - DAM	11.21	32.62	21.68	5.98	1.48
Both - DAM	11.40	24.78	11.67	5.70	4.10
OPT - DAM	11.21	26.69	13.67	**7.13**	2.99
SAR -DAM	**11.43**	**34.02**	**21.94**	6.03	**1.43**

accuracy by 1.80% compared to the No-DAM, 20.20% compared to the Both-DAM, and 15.00% compared to the OPT-DAM. Compared with No-DAM, Both-DAM as well as OPT-DAM decreases the average detection accuracy OA, and only SAR-DAM improves the accuracy by 1.80%, so it can be verified the conjecture that the missing detail information in the proposed network structure comes from the SAR images, and adding detail in the SAR images Attention module can better extract the detail features of SAR image, which makes the fusion classification effect better.

Table X shows the results of different evaluation indexes of adding the DAM. It can be seen that after SAR-DAM, it achieves the highest value in entropy, SD, SF, and the lowest value in MSE, which all prove that the fusion effect is better after SAR-DAM.

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TABLE XI
DETECTION OVERALL ACCURACIES FOR DIFFERENT CATEGORIES OF ADDING ATTENTION MODULES

OA	Forest	Field	Water	Building	Road	Others
No - DAM	0.1375	0.8351	0.6278	**0.6630**	0.5966	0.0069
Both - DAM	**0.3756**	0.6373	0.3364	0.5769	**0.6443**	0.0225
OPT - DAM	0.1990	0.6614	0.5631	0.6079	0.6270	**0.1827**
SAR - DAM	0.0147	**0.9070**	**0.6631**	0.4126	0.1984	0.0549

TABLE XII
KAPPA COEFFICIENTS FOR DIFFERENT CATEGORIES OF ADDING ATTENTION MODULES

KAPPA	Forest	Field	Water	Building	Road	Others
No - DAM	**0.0599**	0.7613	0.4485	**0.4408**	0.2659	0.0018
Both - DAM	0.0281	0.5861	0.3128	0.3410	**0.4435**	0.0090
OPT - DAM	0.0301	0.6255	0.4462	0.3945	0.2721	0.0118
SAR - DAM	0.0119	**0.7799**	**0.4651**	0.3591	0.1715	**0.0123**

In terms of entropy, on the basis of No-DAM, after Both-DAM, it improves by 0.19, OPT-DAM is no improvement, and SAR-DAM improves by 0.22, so the fusion improves the amount of information coming from the SAR image. In terms of SD and SF, on the basis of No-DAM, the Both-DAM not only did not improve the image clarity, but also decreased, and SAR-DAM improved by 1.40 and 0.26, respectively, and the image quality was greatly improved. The MSE, on the basis of No-DAM, Both-DAM, and OPT-DAM increased by 2.77 and 2.02 times, which means that a large number of outliers appeared, and SAR-DAM improved it by 0.03, with better detection and classification. The PSNR after SAR-DAM is 0.05 higher The PSNR after SAR-DAM is 0.05 higher than the effect of No-DAM, and 0.33 higher than Both-DAM. To summarize the above, adding DAM in SAR image can better extract the detail features of SAR image, which makes the fusion effect better.

Table XI shows the detection accuracies of different categories with the addition of the attention module, and Table XII shows the KAPPA coefficients of different categories with the addition of the attention module, the proposed network structure achieves higher values in different categories, which proves that the proposed network structure is more effective in the classification and detection of the proposed network structure. In the field category, the average accuracy of the proposed method is improved by 7.19% than No-DAM, 26.97% than Both-DAM, and 24.56% than OPT-DAM. In the category of water, the average accuracy of the proposed method is improved by 3.53% than No-DAM, 32.67% than Both-DAM, and 10.00% than OPT-DAM. Others category, the KAPPA coefficient of the proposed method is 6.80 times higher than No-DAM, 1.37 times higher than Both-DAM, and 1.04 times higher than OPT-DAM. The remaining building and road categories are analyzed as a sample imbalance problem due to too few labels, which is verified in subsequent experiments.

3) Loss Function: In this section, experiments continue on the WHU-OPT-SAR dataset. Addressing concerns about the model's relatively low SD and PSNR, a novel loss function is introduced, comprising three components: information loss (L_{I}), pixel loss (L_{P}), and noise loss (L_{N}). Pixel loss addresses the SD issue, while noise loss targets the PSNR problem. By incorporating information loss, the fusion detection and classification performance can be enhanced in terms of both SD and PSNR, while preserving fusion information. In addition, several existing image fusion loss functions are examined and integrated into the proposed network structure. Recent studies in various fields have contributed valuable insights. For instance, in land classification research [44], Information loss L_{I} between the fused image and the authentic image is utilized to effectively retain information. Similarly, in multimodal medical image fusion [51], techniques like reconstruction loss L_{R} and feature decomposition loss L_{FD}, derived from transformer-based decoding correlation coefficients, are employed to enhance fusion quality. Furthermore, in the context of unsupervised fusion of optical and SAR images [52] texture loss L_{T} for optical images and structure loss L_{S} for SAR images are proposed to mitigate decomposition loss, a significant factor in medical image fusion. These methods effectively address decomposition-related losses. These newly proposed loss functions are integrated into the proposed network and compared with the novel loss function. Experimental results demonstrate the superior performance of the proposed new version of the loss function.

The findings from Fig. 7 suggest that the proposed loss function significantly enhances image classification research. Upon examining the labeled regions in the figure, it becomes apparent that the classification results obtained with the proposed loss function closely align with the actual labeled categories, unlike the outcomes of other loss functions, which exhibit various classification errors and failures. In particular, Fig. 7(d) illustrates the consequences of solely relying on L_{I}. Here, noise interference obscures accurate identification and classification within the labeled regions. For instance, in the first row, a substantial portion of forest remains incorrectly identified. Similarly, in the second row, regions designated as road and water are only partially recognized, attributed to noise interference that corrupts feature extraction, leading to misclassification as adjacent water bodies. Fig. 7(e) illustrates the consequences of L_{R} and L_{FD}, which missing pixels further

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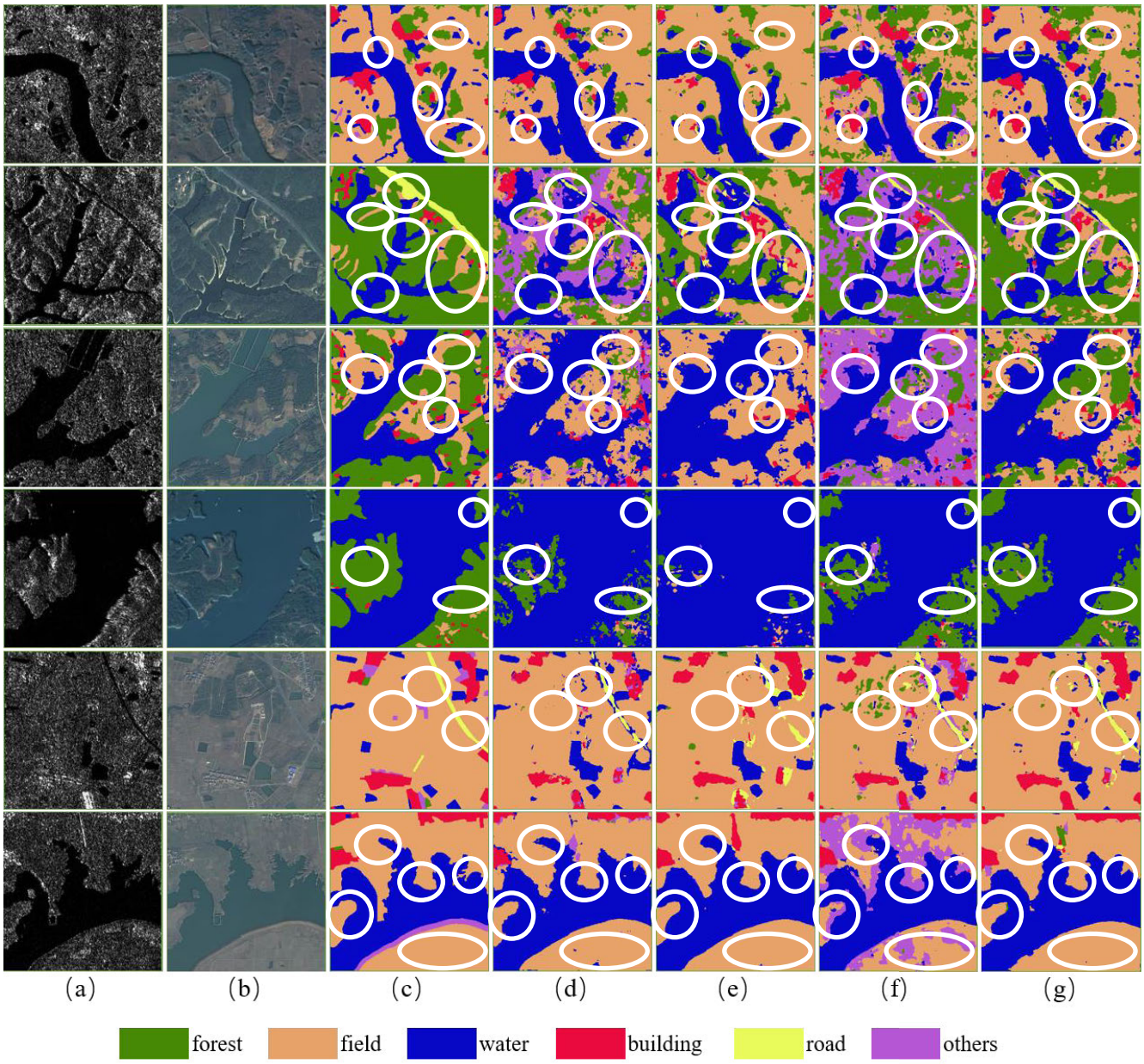


Fig. 7. Classification results by adding loss function fusion detection. (a) SAR images. (b) Optical images. (c) Labeled images. (d) Information loss L_{I} classification results. (e) Reconstruction loss L_{R} and feature decomposition loss L_{FD} classification results. (f) Texture loss L_{T} and structure loss L_{S} classification results. (g) Information loss L_{I}, pixel loss L_{P}, and noise loss L_{N} (ours) classification results.

TABLE XIII
TEST RESULTS OF VARIOUS LOSS FUNCTIONS

Loss Function	OA	IOU	KAPPA
L_I	0.777	0.300	0.509
L_R, L_{FD}	0.761	0.268	0.445
L_T, L_S	0.626	0.286	0.369
L_I, L_P, L_N	**0.789**	**0.337**	**0.525**

TABLE XIV
RESULTS OF DIFFERENT EVALUATION INDICATORS FOR VARIOUS LOSS FUNCTIONS

Loss Function	EN	SD	SF	PSNR	MSE
L_I	11.43	**34.02**	**21.94**	6.03	1.43
L_R, L_{FD}	11.39	28.83	14.52	5.59	1.39
L_T, L_S	11.38	29.64	18.25	6.11	2.36
L_I, L_P, L_N	**11.51**	29.47	16.99	**6.21**	**1.37**

exacerbate the problem, with extensive forest areas erroneously classified as water bodies in the third and fourth rows due to pixel loss. Fig. 7(f) illustrates the consequences of L_{T} and L_{S}, and the absence of critical information results in the misclassification of field regions as forest and road in the fifth row, while in the sixth row, the failure to recognize field areas leads to classification as others. These shortcomings stem from information loss during the fusion process, undermining

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TABLE XV
TEST ACCURACY FOR DIFFERENT CATEGORIES

OA	Forest	Field	Water	Building	Road	Others
L_I	0.0147	0.9070	**0.6631**	0.4126	0.1984	0.0549
L_R, L_{FD}	0.0783	0.8788	0.5054	**0.5804**	0.1526	0.0082
L_T, L_S	0.1188	0.7081	0.5749	0.3652	0.2435	**0.2993**
L_I, L_P, L_N	**0.1988**	**0.9266**	0.6052	0.3972	**0.4033**	0.0674

TABLE XVI
KAPPA COEFFICIENTS FOR DIFFERENT CATEGORIES

KAPPA	Forest	Field	Water	Building	Road	Others
L_I	0.0119	0.7799	0.4651	**0.3591**	0.1715	0.0123
L_R, L_{FD}	**0.0585**	0.7744	0.4028	0.2689	0.1034	0.0061
L_T, L_S	0.0424	0.6523	0.4718	0.3280	0.2148	0.0072
L_I, L_P, L_N	0.0553	**0.7908**	**0.4752**	0.3419	**0.3340**	**0.0222**

fusion effectiveness. In summary, the proposed loss function, encompassing weighted L_{I}, L_{P}, and L_{N}, yields superior fusion detection classification compared to alternative approaches, as evidenced by qualitative analysis of the marked areas in the figure.

Table XIII shows the results of fusion detection by adding various loss functions, and it can be seen that, comparing with other loss functions, the proposed new loss function has the highest test OA, IoU, and KAPPA coefficient, and the fusion effect is better than other loss functions. In terms of average accuracy OA, the proposed loss function improves 1.20% over L_{I}, 2.80% over L_{R} and L_{FD}, and 16.30% over L_{T} and L_{S}. In terms of IoU, the proposed loss function improves 3.70% over L_{I}, 6.90% over L_{R} and L_{FD}, and 5.10% over L_T and L_{S}. In terms of KAPPA coefficients, the proposed loss function improves 1.60% over L_{I}, 8.00% over L_{R} and L_{FD}, and 15.60% over L_{T} and L_{S}. Taken together, the fusion effect of the proposed loss function is much higher than other loss functions. The network structure control tests are all using L_{I}, where it is also verified that the proposed new loss function, which also has a great improvement on the basis of L_{I}, also plays an important role in the quality of the overall image fusion.

Table XIV shows the results of different evaluation indexes of fusion detection by adding various loss functions. When the loss function is L_{I}, the SD and SF are slightly higher than the proposed loss function, but the entropy and PSNR of the proposed loss function are higher than the other loss functions, and the MSE is much lower than the other loss functions, so that the fusion effect is much better than the other loss functions only in terms of evaluation indexes. Fusion effect is much better than other loss functions only in terms of evaluation indexes. In the entropy, the proposed method improves 0.08 over the L_{I}, 0.12 over the L_{R} and L_{FD}, and 0.13 over the L_{T} and L_{S}, which indicates that the addition of weighted L_{I} improves the amount of information a lot. In the PSNR, the proposed method improves 0.18 over the L_{I}, 0.62 over the L_{R} and L_{FD}, and 0.10 over the L_{T} and L_{S}, which indicates that the addition of noise loss improves the amount of information a lot. 0.10, indicating that after adding the noise loss, the noise of the fused image is reduced a lot, and the quality of image fusion is better. In the MSE, the proposed method is lower than the L_{I} by 0.06, than the L_{R} and L_{FD} by 0.02, than the L_{T} and L_{S} by 0.99, which indicates that the proposed method fuses the least outliers, and is the closest to the real situation. The spatial frequencies are all the highest when there is only L_{I}, and in order to reduce the impact of sample imbalance, weights are added to the L_{I}, so there is a decrease in some evaluation indexes, but the overall fusion effect is still better than other loss functions.

Table XV shows the detection OA and Table XVI shows the KAPPA coefficients of different categories with the loss function, the proposed loss function achieves higher values in different categories, which prove that the proposed loss function is more effective in classification and detection. In forest category, the OA of the proposed method is 18.41% higher than the L_{I}, 12.05% higher than the L_{R} and L_{FD}, and 8.00% higher than L_{T} and L_{S}. Field category, the proposed method has a high average accuracy of 92.66%, which is 1.96% higher than the L_{I}, 4.78% higher than the L_{R} and L_{FD}, and 21.85% higher than L_{T} and L_{S}. Water category, the proposed method has a higher KAPPA coefficient than the L_{I} by 1.01%, L_{R} and L_{FD} by 7.24%, L_{T} and L_{S} by 0.34%. In the category of road, the OA of the proposed method is 20.49% higher than the L_{I}, 25.07% higher than the L_{R} and L_{FD}, and 15.98% higher than the L_{T} and L_{S}. In the category of others, the KAPPA coefficient of the proposed method is 1.80 times higher than that of L_{I}, 3.64 times higher than that of L_{R} and L_{FD}, and 3.08 times higher than that of L_{T} and L_{S}. And the building category, although not the highest, is just 1.72% lower than the KAPPA coefficient of L_{I}, 7.30% higher than that of L_{R} and L_{FD}, and 1.39% higher than that of L_{T} and L_{S}. Comprehensively, it can be seen that the proposed loss function with the addition of weights effectively mitigates the problem of classification errors caused by sample imbalance, and compared with several of the most typical loss functions in

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recent years, the proposed method of fusion detection has the best classification effect and the highest classification quality.

IV. Conclusion

In this article, a dual-encoder network structure (DEN) for fusion of SAR images and optical images was proposed to effectively deal with the noise interference and sample imbalance problems encountered by traditional methods in the fusion process. By introducing a noise suppression module (DAM) and a new weighted loss function, the experimental results showed that the proposed method not only outperformed the existing methods in terms of fusion quality, but also significantly improved the classification accuracy and model complexity. The proposed dual-encoder structure made full use of the respective advantages of SAR and optical images, and significantly reduced the noise impact by deep fusion at the feature level, while ensuring the image details and texture information. Through effective module design and loss function optimization, the complexity of the model complexity was improved by 15.3%, while the OA was improved by 2.6%, and the entropy (EN), PSNR, and MSE were improved by 29%, 27%, and 7%, respectively.

Despite the excellent performance of the current experimental dataset, the ability of the method to generalize to different environments and sensor conditions requires further validation. Future study could consider testing on more diverse datasets to validate the model's adaptability. In addition, despite the reduction in model complexity, real-time performance in large-scale, high-resolution data processing remains a challenge. Future research can further optimize the algorithm to improve the real-time processing capability. Overall, the DEN method proposed in this article has improved the fusion quality and classification performance, however, in practical applications, data diversity and processing efficiency need to be considered, which will be a key direction for future research.

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